ORIGINALS · PRINTED HERE · DO NOT CHANGE

Small edits. *Different worlds.*

Two real specifications. Change one word at a time. Does the meaning survive?

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ABSTRACT

Semantic drift is the gap that opens when a specification is rewritten and the words still look right but the set of behaviours they permit has moved. Every verified system has two links: the system satisfies the specification, and the specification says what we meant. Provers check the first. Nothing routinely checks the second, and specifications are revised constantly: translated into formal notation, clarified by reviewers, rewritten by models, edited one word at a time in public constitutions and model specs.

For AI safety this is the weak joint. Oversight, evaluation and control all read a specification and ask whether the system complied. If the specification drifted first, compliance proves the wrong thing. The companion paper measures drift with a certified oracle on 500 algebraic laws. This poster asks visitors to produce it, one word at a time, on two real specifications, and to judge whether each word kept the meaning.

THE PROBLEM

$P \models \varphi$
Verification. Does the system satisfy the spec? Provers do this.

$\varphi \approx \text{intent}$
Validation. Does the spec still say what we meant? Nobody does this, and specs get revised all the time.

WHAT IS MEASURED

textual How many words differ from the original. The number in the top bar of the page.

crowd For every change, how many visitors judged it `same` `changed` `unresolved`, before and after seeing the dictionary.

There is no oracle for whether "flourishing" means "thriving". Nobody is told who is right. Disagreement is the result, not noise, and it is recorded per word.

HOW THE CHAIN MOVES

change Tap a word. A dictionary sheet shows its definition and up to eight alternatives with theirs. Pick one. One word per change.

judge Other visitors see the sentence before and after. Same meaning or changed? Commit, read both definitions, decide again.

chain Visitors change the *latest* text, not the original. The paper keeps the originals; the screen shows what the swaps have made of them.

THE ORIGINALS · FIVE SENTENCES EACH, ASSEMBLED FROM QUOTATIONS · PRINTED HERE, REWRITTEN THERE

ANTHROPIC · CLAUDE'S CONSTITUTION

1. "We want Claude to be exceptionally helpful while also being honest, thoughtful, and caring about the world."
2. Claude should be broadly safe, broadly ethical, compliant with Anthropic's guidelines, and genuinely helpful, and "in cases of apparent conflict, Claude should generally prioritize these properties in the order in which they are listed."
3. Broadly safe means "not undermining appropriate human mechanisms to oversee AI during the current phase of development," including "the ability of legitimate principals to adjust, correct, retrain, or shut down AI systems."
4. "Claude should basically never directly lie or actively deceive anyone it's interacting with," and should be "diplomatically honest rather than dishonestly diplomatic."
5. "We want Claude's helpfulness to flow from deep and genuine care for users' overall flourishing," on "a path towards mutual flourishing for both AI and humanity."

OPENAI · MODEL SPEC, 29 OCTOBER 2026

1. "Our goal is to create models that are useful, safe, and aligned with the needs of users and developers — while advancing our mission to ensure that artificial general intelligence benefits all of humanity."
2. "Humanity should be in control of how AI is used and how AI behaviors are shaped."
3. Instructions carry different levels of authority — Root, System, Developer, User, Guideline — and "higher-authority instructions take precedence over lower-authority ones."
4. "Represent facts, evidence, uncertainty, capabilities, and actions truthfully."
5. OpenAI aims to "prevent our models from causing serious harm to users or others," and the model "must not optimize for or directly pursue any additional goals as ends in themselves."

A TRAIL OF REASONABLE-LOOKING SWAPS

WORDING	WHAT THE DICTIONARY SAYS
"...care for users' overall **flourishing**"	flourishing · growing, thriving, prospering
↓	↓
"...care for users' overall **thriving**"	thriving · that thrives; successful
↓	↓
"...care for users' overall **prosperity**"	prosperity · good fortune and wealth

Each step is a dictionary synonym of the one before.

Two synonym steps. The care is now about money.

At which word did *the intent* change?

Illustrative. The page records the real swaps and what the crowd made of each.

Judge a change. *Then change a word yourself.*

Judge a change. One word moved. Same meaning or changed? Commit, read both definitions, decide again, see how others judged. Four rounds, two minutes.

Change a word. Tap it, see what it means, pick an alternative with its definition. It becomes the current text for everyone.

The drift so far. Every swap in order with the crowd's verdict on each, for both specifications.

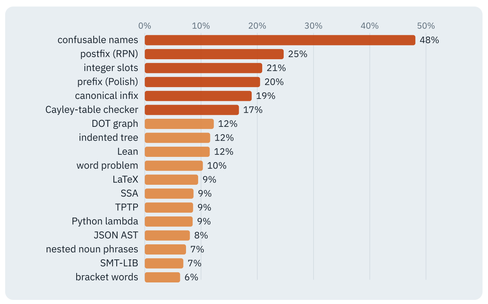
The formal companion. A compute-budget rule where an oracle checks what each one-token edit does.

https://corpaci.com/fork-this-spec/

FOR THE CONVERSATION

WHAT THE COMPANION PAPER MEASURED

The same 500 magma identities, written in 18 notations, read by nine models and written back. A certified implication oracle labels each output equivalent, weaker, stronger or incomparable. Bars: share of certified outputs that are **provably weaker** than the law the model was given.



Weakening is not noise. Under confusable identifiers 93.7% of weaker outputs merged two variables into one; under Latin/Cyrillic homoglyphs 81% of outputs were weaker and 87% of merged pairs were look-alikes. Certified recovery runs from 73.7% (bracket words) to 8.5% (postfix). Selecting for validity moved failures from loud to silent without improving survival. Here the dictionary is the selector: every swap is a certified synonym.

JUDGE A CHANGE

A sentence before and after one word moved. **Same meaning or changed?** Commit, then the dictionary shows both definitions. Decide again. Then see how others judged the same swap. Nobody is told who is right.

Tallied per swap: same, changed, unresolved, and how often the definitions changed a mind. **Exact equivalence is transitive:** if every swap had preserved meaning, the chain could not drift. Where the crowd splits, the word was carrying more than one meaning.

WHERE THESE SPECIFICATIONS SIT

Written constitutions and model specs are one of five families of answers to "aligned to what". The others: assistance games, where the agent stays uncertain about human objectives (Russell et al., CIRL); approval-directed agents and amplified oversight (Christiano); corrigibility, staying correctable against the current objective (Soares, Fallenstein, Yudkowsky, Armstrong); and collective, democratically informed alignment (Collective Constitutional AI). Coherent extrapolated volition is the provocative comparison. Each is a text somebody will revise.

ASK ME ABOUT

1. Iterated revision with a reviewer is a refinement process. Does it converge, and to whose intent?
2. Which words in a constitution should never be allowed a synonym, and who decides?
3. Can a revision step be typed as a lens whose backward map carries the preservation certificate?

STICK A DOT ON THE SENTENCE YOU WOULD CHANGE FIRST

1	2	3	4	5	none
sentence					leave it

Anthropic · Claude's constitution

OpenAI · Model Spec

Green dot if you believe the word you have in mind has a synonym that keeps the meaning. Orange if you believe any change to that sentence moves it. Write the word next to the dot.

REFERENCES

The two specifications
· Anthropic (2026). *Claude's constitution.* anthropic.com/constitution · OpenAI (2026). *Model Spec*, 29 Oct 2026. model-spec.openai.com · Bai et al. (2022). Constitutional AI. arXiv:2212.08073 · Anthropic (2023). Collective Constitutional AI.

Measuring drift
· Patel & Corpaci (2026). *When Specifications Drift.* NeurIPS VeriCodeGen · Dalrymple et al. (2024). Towards Guaranteed Safe AI. arXiv:2405.06624 · Goguen & Burstall (1992). Institutions. JACM · Fonseca et al. (2017). Correctness of formally verified distributed systems. EuroSys.

Intent, requirement, specification
· Jackson (1995). The world and the machine. ICSE · Zave & Jackson (1997). Four dark corners of requirements engineering. TOSEM · Krakovna et al. (2020). Specification gaming. DeepMind · Manheim & Garrabrant (2018). Variants of Goodhart's law. arXiv:1803.04585 · Hadfield-Menell et al. (2017). Inverse reward design. arXiv:1711.02827.

Alignment targets
· Hadfield-Menell et al. (2016). Cooperative inverse RL. arXiv:1606.03137 · Christiano (2014). Approval-directed agents · Soares, Fallenstein, Yudkowsky & Armstrong (2015). Corrigibility. AAAI-WS · Shovelain & Skalse (2022). Updating utility functions. Alignment Forum · Yudkowsky (2004). Coherent extrapolated volition.

Scalable oversight and its problems
· Irving, Christiano & Amodei (2018). AI safety via debate. arXiv:1805.00899 · Christiano, Shlegeris & Amodei (2018). Iterated amplification. arXiv:1810.08575 · Bowman et al. (2022). Measuring progress on scalable oversight. arXiv:2211.03540 · Burns et al. (2023). Weak-to-strong generalization. arXiv:2312.09390 · Khan et al. (2024). Debating with more persuasive LLMs. arXiv:2402.06782 · Kenton et al. (2024). Weak LLMs judging strong LLMs. arXiv:2407.04622 · Barnes & Christiano (2020). Obfuscated arguments · METR & Redwood (2026). OpenAI/Hugging Face incident investigation.

The paper preserves the originals. The screen reveals what one-word changes have made of them. Every change is kept with its author or model, the word pair, both definitions, the stated reason and the crowd's verdict. The chain is published after the event.

Bring a word.
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